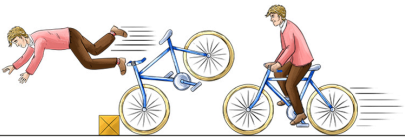
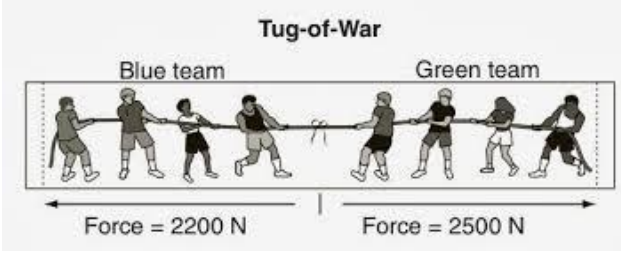
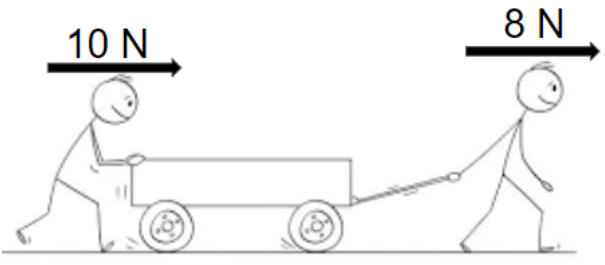


Name _____ HR _____



8th Grade Science Unit 1 Changing Motion STAR Cards!

QUESTION	ANSWER
1a) ☆☆☆ 1) If a man pushes off a boat as he jumps to the land, what would happen to the boat?  2) This is an example of Newton's <u>3rd Law of Motion</u>. Describe <u>Newton's 3rd Law</u>.	1a) - The boat would be pushed backwards. (*The opposite direction of the man) - For every action, there is an equal and opposite reaction
1b) ☆☆☆ Based on <u>Newton's 3rd Law</u>, if the car and truck below collide, which vehicle will exert more force? 	1b) Both vehicles will exert the same force!! (Newton's 3rd Law says that for every action, there is an <u>equal</u> and opposite reaction. - this means when two things collide, they exert equal and opposite forces on each other.)
1c) ☆☆☆ 1) Which requires more force to move - a baby stroller or a car?  2) <u>Newton's 2nd Law</u> explains why this is true. Explain Newton's 2nd Law.	1c) - The Car <u>Newton's 2nd law</u> states that the more mass an object has the more force is needed to move it (accelerate it).

QUESTION	ANSWER
1d)  What is the formula that calculates the “Force” in <u>Newton’s 2nd Law</u>?	1d) Force = Mass x Acceleration
1e)  1) What happens to a bike rider who hits a big rock (or other obstacle) ?  2) This is an example of Newton’s 1st Law of Motion. Explain <u>Newton’s 1st Law of Motion</u>.	1e) 1) The rider would continue to move forward. 2) An object will continue to move in the same direction and the same speed - unless an unbalanced force acts on it.
1f)  1) Explain which team will win this tug-of-war  2) What is the net force of this action? Why?	1f) 1) The Green team will win because it is applying greater force. 2) The net force will be 300 Newtons, because when forces are going in opposite directions, you subtract them. (2500 - 2200 = 300.)
1g)  1. In which direction will the wagon move? 2. What will the net force be and why? 	1g) 1. The wagon will move to the right 2. The net force will be 18 Newtons because when forces are going in the same direction, you add them (10+8=18)